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A 250 mV $Cu/SiO_2/W$ Memristor with Half-Integer Quantum Conductance States

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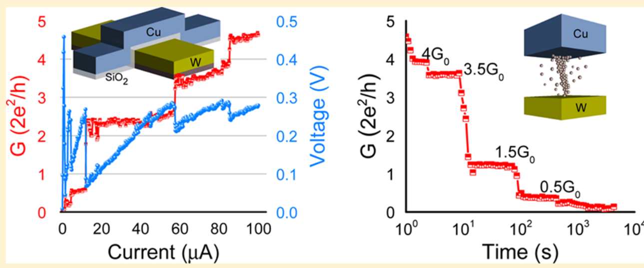
Supporting Information

ABSTRACT: Memristive devices, whose conductance depends on previous programming history, are of significant interest for building nonvolatile memory and brain-inspired computing systems. Here, we report half-integer quantized conductance transitions $G = (n/2)\ (2e^2/h)$ for $n = 1, 2, 3$, etc., in $Cu/SiO_2/W$ memristive devices observed below 300 mV at room temperature. This is attributed to the nanoscale filamentary nature of Cu conductance pathways formed inside SiO_2. Retention measurements also show spontaneous filament decay with quantized conductance levels. Numerical simulations shed light into the dynamics underlying the data retention loss mechanisms and provide new insights into the nanoscale physics of memristive devices and trade-offs involved in engineering them for computational applications.



KEYWORDS: *Memristor, half-integer quantization, filamentary conduction, nanoelectronics, resistive switching, Cu, SiO_2*

Memristor is a fundamental circuit element whose resistance states depend on its past operating history and was predicted by Leon Chua in 1971 based on symmetry arguments.[1] Recently, memristive characteristics have been observed in two terminal metal−insulator−metal (MIM) devices whose conductivity is modified by the electric field driven atomic rearrangement of mobile charged ions within the insulator.[2−5] These devices are being extensively studied for building next-generation memory technologies as well as for non-Von Neumann computing systems inspired by the brain.[6−9] One of the main attractions of these devices is its simple structure, enabling high integration densities at nanoscale dimensions. However, significant challenges specifically related to device-to-device variability, scalability, state stability, and optimization of switching power and speed have to be addressed before commercially viable technologies can be manufactured based on these devices.[3,10−12] Also, a coherent understanding of the underlying physics of memory switching in these devices has not been well-established so far.

It is believed that one of the main reasons for the large variability in operating characteristics of these devices is due to the inherent nonuniformity associated with amorphous dielectric films used to create the MIM structure. This could result in electric field localization at different regions within the dielectric when a programming voltage is applied to the device, creating filamentary conduction pathways.[11,13−16] The conductance of nanoscale devices could exhibit discrete quantized states when the lateral dimension of the conduction pathway is comparable to the Fermi wavelength (λ_F) and the transverse length is less than the mean-free path (l). Landauer theory for ballistic electron transport predicts $G_0 = 2e^2/h$ as the quantum of conductance where e is the charge of electron and h is the Planck's constant and the factor 2 accounts for spin degeneracy.[17,18]

The constraint to be compatible with the current CMOS technology is a main driving factor in the material choices for logic and memory devices for the future. A SiO_2-based device with Cu top electrode is a suitable candidate for the low temperature back-end-of-line (BEOL) process integration. Also, Cu is known to be highly mobile in SiO_2. The high mobility of ionic species in the dielectric is favorable for low power operation, though may have deleterious effects on the data retention capabilities.[19] Our experiments and analysis are designed to understand the fundamental mechanisms of conductance switching in these devices, scaling limits, and directions for the material trade-offs.

Our devices consist of a SiO_2 dielectric layer sandwiched between an active Cu top electrode and an inert Ti/TiN/W bottom electrode, over a p-type Si(100) wafer with a SiO_2 electrical isolation layer. The Ti/TiN/W stack used as a bottom electrode was sputter-deposited and patterned with a photolithography and lift-off process. The 10 nm thick SiO_2 film and the Cu top electrode were then sputter-deposited over a patterned double layer photoresist (LOR3B and S1813) in a single vacuum and room temperature process. The 100×100 μm^2 cross-point devices (Figure 1a) were finally obtained by a common lift-off process. The stoichiometry of the SiO_2 layer was confirmed by X-ray photoelectron spectroscopy on the Si

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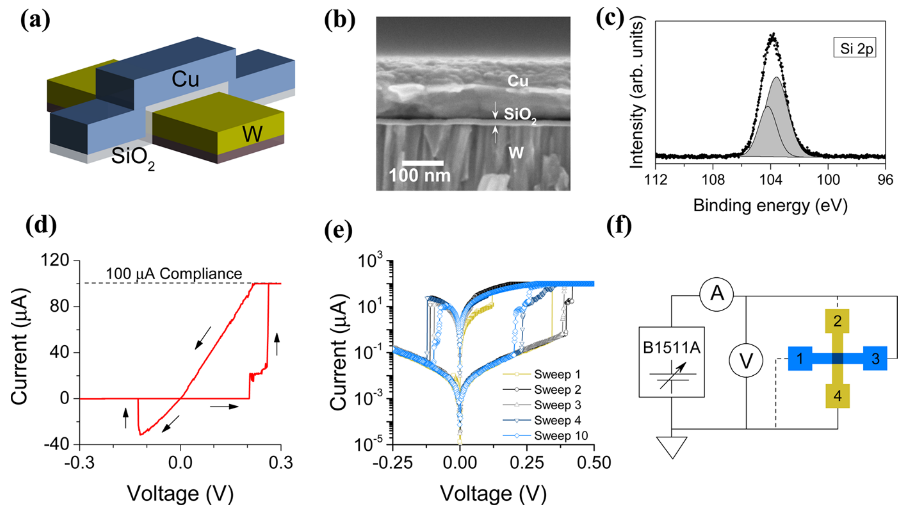

Figure 1. (a) Crossbar structure of the fabricated device. (b) FEGSEM (field emission gun scanning electron microscope) image of the device (cross section) showing 10 nm SiO_2 sandwiched between Cu and W. (c) Si 2p XPS spectrum of a 10 nm sputter-deposited SiO_2 film. Our quantitative analysis suggests O/Si ratio of 2. (d) Typical low voltage pinched hysteresis observed in the device in linear scale. (e) The first four and the tenth sweep across the device are shown in log scale. (f) Electrical setup schematic used to characterize the device at the cross-point junction is shown. The connection via device contact pads 1 and 2 or 3 and 4 are used for the device characterization, and those through 1 and 3 or 2 and 4 are used for the electrode characterization.

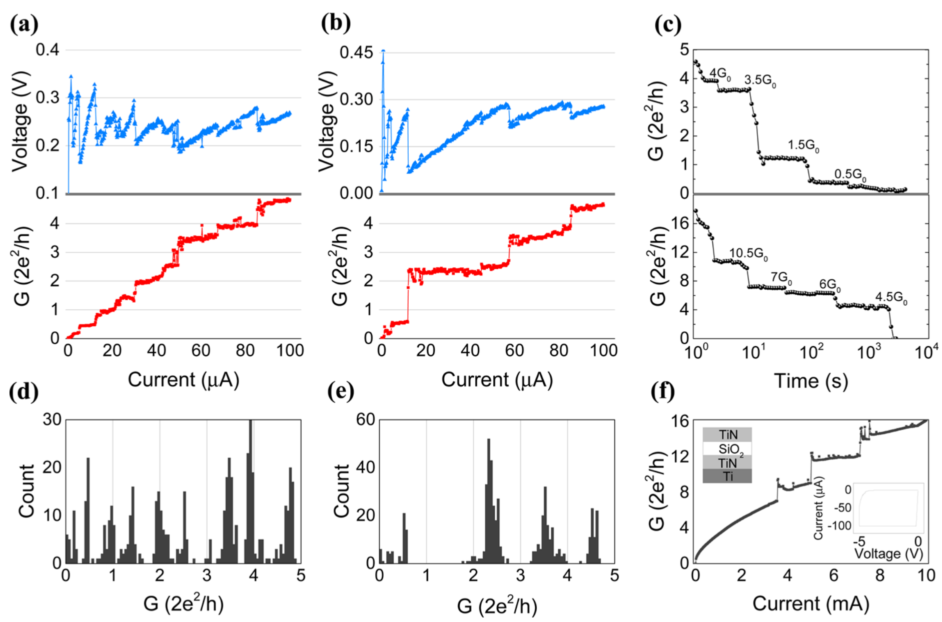

Figure 2. (a,b) Voltage across the device and corresponding conductance change of two Cu/SiO_2/W based devices during current sweep measurements. (c) Conductance decay as a function of time due to the Cu filament degradation inside the SiO_2 dielectric for two different initial quantized levels. (d,e) The corresponding conductance histogram of the devices quantization levels shown in (a,b). Here, the plot in a shows all the possible quantized levels, while the plot in b shows only fractional multiples of G_0. We observe that the first voltage drop and the histogram indicate the presence of a quantized conductance level below 0.5 G_0. (f) Quantized conductance measurement across TiN/SiO_2/TiN/Ti device observed at high voltage and currents. The $I-V$ plot in the inset shows the forming process, which happens at much higher voltages than the devices having Cu as the top electrode.

2p core level (contribution of Si $2p_{3/2}$ at 103.6 eV, Figure 1c).[20−22] The devices were subjected to 400 °C, 5 min annealing in Ar environment after top electrode patterning.

The device exhibit bipolar resistance switching with a pinched hysteresis loop characteristic of memristive switching, as in Figure 1d. During positive sweep, the devices undergo SET transition where it switches from HRS (high resistance state) to LRS (low resistance state), and the final resistance is determined by the magnitude of the current compliance imposed by the measurement circuitry. In the negative sweep direction, the devices are RESET at about −100 mV. The freshly fabricated and the cycled devices showed no appreciable difference in its high resistance levels. Although the SET switching threshold of the devices gradually reduce from 0.4 to 0.2 V within the first few cycles, the devices follow the same HRS path (Figure 1e). This suggests that the conduction pathways are completely removed during the reset process.

Note that there are sudden multilevel conductance transitions during the SET sweep in Figure 1d. To understand this better, we used a current sweep (200 nA/10 ms)

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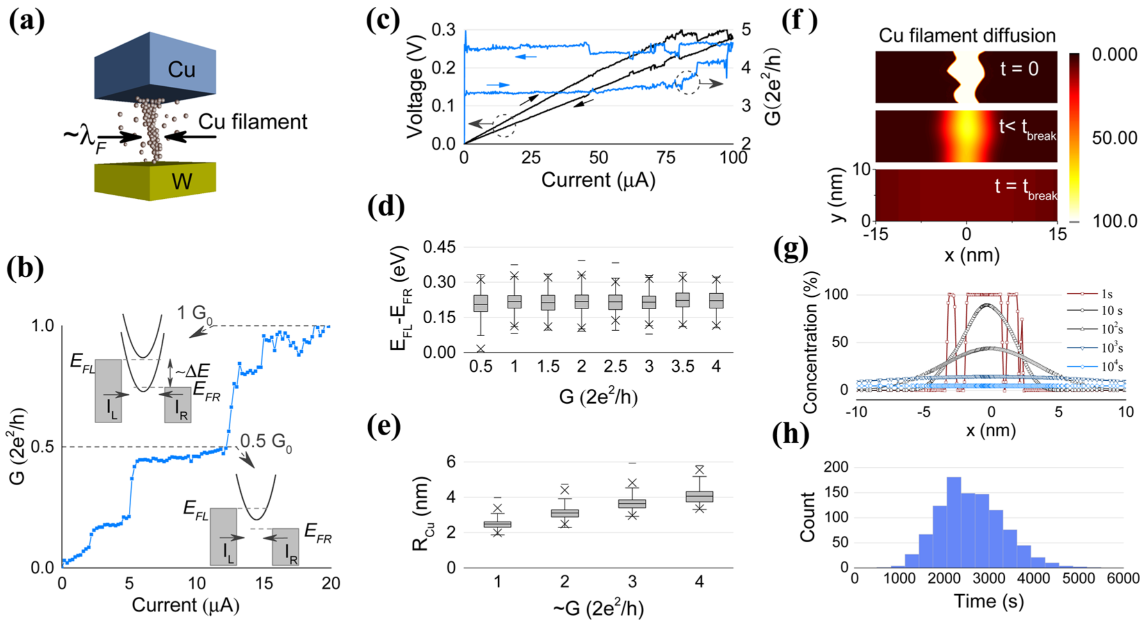

Figure 3. (a) Schematic of Cu filament formed inside the SiO_2 dielectric. (b) $E-k$ band representation for 0.5 G_0 and 1 G_0 states marked on the corresponding G vs I plot. (c) Dual current sweep measurement across an already formed filament which shows a conductance transition from 3.5 G_0 to 4.5 G_0. (d) Plot of the Fermi level splitting ($E_{FL}-E_{FR} \sim$ eV) across the filament in each quantized conductance state ($(n/2)G_0 \pm 0.05\ G_0$) during filament formation in a current sweep. (e) An estimate of Cu filament radius (R_{Cu}) using eq 2. (f) Fick's law simulation of Cu diffusion in our devices: An example Cu filament profile in SiO_2 at $t = 0$, $t < t_{break}$, and $t = t_{break}$, where t_{break} is the filament break time. The three plots have a common color scale. (g) Filament decay as a function of time due to diffusion of Cu filament structure in SiO_2 dielectric. (h) Distribution of filament break times for various initial Cu filament profiles.

measurement in the range of 0−100 μA where the voltage across the devices are monitored. We observed conductance quantization levels at integer multiples and, rather unexpectedly, also at half-integer multiples of G_0 (Figure 2a,b). This quantized switching phenomenon was observed in multiple devices in several current sweep measurements consistently, although some conductance levels were missing occasionally (Figure 2b).

The device conductance was also found to spontaneously decay from its LRS to HRS. To probe this further, the devices were programmed to LRS by applying a triangular voltage waveform with a peak amplitude of 0.5 V and a duration of 2 s, which pushed the devices into conductance states in the range of 5−20 G_0. The conductance of the devices were then monitored at logarithmically distributed intervals using a 50 mV voltage pulse lasting 5 ms. Half-integer quantized levels were observed during this spontaneous decay process as well and the filaments completely decayed within 10^4 s in multiple devices as shown in Figure 2c. Some possible quantized levels were missing, which may be due to the nonuniform filament degradation process or prolonged time interval between successive measurements.

The quantization levels observed during current sweep measurements and the spontaneous decay process suggest that the resistance switching is due to localized nanoscale conduction pathways formed between top and bottom electrodes in our devices. To study the origin of these pathways, we fabricated a control device where the 10 nm sputter-deposited SiO_2 is sandwiched between inert top and bottom electrodes made of TiN. The $I-V$ characteristic of such a $TiN/SiO_2/TiN/Ti$ device is shown in Figure 2f. The freshly fabricated device is in a high resistance state (~500 MΩ) and switches to LRS values only after being subjected to relatively larger forming voltages of amplitude 5 V (figure in inset). These devices also displayed quantized conductance, but at much higher current (several mAs) and voltage levels (above 3 V). Similar behavior has been observed earlier and was attributed to oxygen-vacancy-based filaments.[15,23,24] Hence, it is clear that Cu ions from the active top electrode play a significant role in the low voltage quantized conductance states observed in our devices.[25]

In order to rule out any interfacial origin for the quantized switching seen in our devices, we fabricated material stacks consisting of just the top and bottom electrodes without the SiO_2 dielectric. A linear $I-V$ characteristic was obtained without any hysteresis, showing that SiO_2 was indeed necessary for the switching transitions (see Supporting Information Figure S2). Hence, we concluded that the origin of quantized conductance switching cannot be due to any field driven modifications at the interfaces or due to possible oxides of tungsten.[26,27]

Though there are numerous reports on Cu/SiO_2 based memristors proposing filamentary switching, conductance quantization which can act as conclusive proof for the filamentary nature has not been reported in this material system to the best of our knowledge.[28−33] Here we observed quantized conductance in Cu/SiO_2 crossbar devices in repeated experiments by current sweep measurements allowing us to conclude that resistive switching in the Cu/SiO_2 system is indeed due to Cu nanofilamentary paths formed inside the SiO_2 dielectric between top and bottom electrodes. However, it is not clear if the quantized conductance can be attributed to single or multiple Cu filaments in the SiO_2 dielectric. It has been shown that the quantized conductance of individual nanofilaments can add together when multiple filaments occur in parallel.[34] Therefore, the missing quantum levels as in Figure 2b might be the result of 0.5 G_0 steps of multiple filaments adding together where a new transport mode is added to

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multiple filamentary channels at the same time. However, in the following discussion and simulations, we assume a single filament for simplicity.

Now we consider the possible mechanisms underlying the observed half-integer quantization ($G = (n/2)G_0$) and filament retention dynamics in our devices. Ballistic electron transport in one-dimensionally confined channels has been of significant interest for nanoscale device engineering. When the transverse dimension of the channel becomes less than the mean free path, electrons barely undergo any collision, except probably at the confining walls, enabling ballistic motion across the channel. If the lateral dimension is comparable to Fermi wavelength of the material, the number of discrete sub-bands available for the electron transport becomes limited.[18,35,36] Confined electron transport has been observed in point contacts in III−V semiconductors and at mechanically controllable break junctions in metal elongation experiments.[37,38] Conductance measurements using low amplitude AC signal in these experiments revealed quantized levels at integer multiples of G_0 when electrons were confined to a one-dimensional filamentary path. Theoretically, each sub-band available for transport in the 1-D channel will contribute a conductance unit of $2e^2/h$.[37,39] Therefore, each conductance step of G_0 corresponds to the addition of a new sub-band to the channel.

However, we observed conductance levels at half-integer multiples of G_0 in addition to the integer quantization levels. Such characteristics have been observed previously in devices where there was a finite voltage, V, applied across the filament, resulting in a finite Fermi level split (= eV) between the two contact reservoirs. This half-integer quantized behavior has been attributed to a situation where the number of sub-bands available in the nanofilament for the current injection from the left and right contacts differs.[23,40−44]

Mathematically, for a single mode nanofilament, where there is only one sub-band available below both the contact Fermi levels, the density of states, n_{1D} obeys the relation: $n_{1D}v_g = (2/h)\theta(E)$ where v_g is the electron velocity and $\theta(E)$ is the Heaviside step function of energy.[45] For multimode ballistic transport with quantized energy levels located at E_i, the current (assuming zero temperature limit) is given as

$$\begin{aligned} I &= I_L - I_R = e\int_{-\infty}^{\infty} dE n_{1D} v_g [f(E - E_{FL}) - f(E - E_{FR})] \\ &= \frac{2e}{h}\int_{E_{FR}}^{E_{FL}} \sum_i \theta(E - E_i) dE \\ &= \frac{2e}{h}\left(\sum_{E_i < E_{FR}} \int_{E_{FR}}^{E_{FL}} \theta(E - E_i) dE \right. \\ &\quad \left. + \sum_{E_{FR} < E_i < E_{FL}} \int_{E_i}^{E_{FL}} \theta(E - E_i) dE \right) \end{aligned} \tag{1}$$

where I_L and I_R are the electron currents injected into the channel from the left and right contacts and E_{FL} and E_{FR} are their Fermi levels, respectively (Figure 3b).

The relative position of quantized energy levels in the nanofilament with respect to the contact Fermi levels is affected by the potential distribution in the filament. Therefore, it is more accurate to use $n_{1D}(E-U)$ than $n_{1D}(E)$ where U is the self-consistent potential, determined by the capacitance and charge inside the nanofilament.[46,47] Assuming reflection-less contacts and scatter free channels, if we have an energy level halfway between E_{FL} and E_{FR},[23,42,44] from eq 1 we get

$$I = \frac{2e}{h} N_R eV + \frac{2e}{h} N \frac{eV}{2} = G_0 N_R V + G_0 \frac{N}{2} V$$

where N_R is the number of sub-bands below E_{FR} and N (= 1 or 0) is the number of sub-bands between the contact Fermi levels. Hence, when $N = 1$, half-integer quantized conductance states could be observed. However, in reality there will be deviations from these ideal conditions, and it will lead to quantized levels to show a distribution around its theoretical values as seen in our device conductance histograms.

Half-integer levels have been reported in a number of valence change memory (VCM) devices, where the agglomeration of oxygen vacancies in a dielectric forms the conduction pathway.[23,48−50] But in these devices, quantization was observed at higher voltages and currents. It has been suggested that half-integer quantization will be difficult to observe in electrochemical metallization cells (ECM) where the species involved in conduction modulation are metal ions, since large potential drops can not be sustained across conductive metal filaments.[23] But such a generalization might not be accurate considering the results from our Cu-based devices and Ag-based devices reported earlier.[51−53] The actual device dielectric may have an effect on the half-integer quantization behavior. For instance, for Ag-based memristors, half-integer quantization in conductance has been reported with SiO_2 dielectric,[52−54] as well as in P3HT:PCBM polymer[51] based ECMs but not with AgI.[53] As seen in Figure 3c, when we performed a current sweep measurement on our device which was programmed to 3.5 G_0 state, a potential drop up to ∼0.3 V was sustained across the filament, before its transitions to 4.5 G_0, and this state is held relatively constant until the voltage across the filament falls below a few millivolts during the reverse current sweep. We believe that the minimum Fermi level split that is needed for a device to show half-integer quantization levels is only limited by the thermal energy of electrons.[55]

In our single filament model, we can estimate the lateral dimension of the nanofilament by measuring the voltage drop across the device during conductance transitions, which is a measure of sub-band spacing. If we neglect the voltage drop across the series contact resistance, these voltages will correspond to the Fermi level splitting of the contacts, ($E_{FL}-E_{FR} \sim$ eV) (Figure 3d). From the sub-band scheme shown for 1 G_0 in Figure 3b, we can see that maximum Fermi level splitting is approximately the sub-band spacing (ΔE) for the discrete energy levels of one-dimensional filament. The distribution of voltage drops across the device when device conductance, $G = (n/2)G_0 \pm 0.05\ G_0$ is displayed in Figure 3, whose mean value for each quantized level remains relatively constant (mean of the average voltage values = 0.216 V). This approximately constant voltage across the filament suggests that the theoretical increase in sub-band spacing is being compensated by the growing filament radius. By solving Schrodinger's equation for a cylindrical conductive filament structure, and using the distribution of sub-band spacing obtained from Figure 3d, we can estimate the radius of the cylindrical filament in our device from the relation:

$$R_{Cu} = \left[\frac{\hbar^2}{2m^*\Delta E}\right]^{1/2} \sqrt{(x_{n+1}^l)^2 - (x_n^l)^2} \tag{2}$$

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where x_n^l is the nth zero of Bessel function to the lth order and m^* (= 1.01 m_e) is the electron effective mass[56] in Cu. Figure 3e shows the estimated Cu filament radius, which has a mean value in the range of 2.5−4.0 nm for the initial quantized levels.

We also observed that the conductance of the Cu filament decays spontaneously in a time-scale of ~10^4 s (Figure 2c). It has been reported earlier that due to the high chemical gradient surrounding the nanofilament, it may react with the surrounding medium or the filament surface may get passivated.[11] However, the negative energy of formation of Cu oxides ($\Delta G^{o}_{f(\mathrm{CuO})}$ = −130 kJ/mol and $\Delta G^{o}_{f(\mathrm{Cu_2O})}$ = −146 kJ/mol) is much lower than that of silicon dioxide (ΔG^{o}_{f} = −856.7 kJ/mol) and, hence in our devices, Cu will most likely be an interstitial element in SiO_2. Therefore, we believe that it is a concentration gradient driven diffusion of Cu atoms into the surrounding SiO_2 that leads to reduction in filament diameter and its eventual dissolution.[32] It has also been reported that the device retention, especially in the off state could be affected by the nonequilibrium "nanobattery effect";[57] however, in our device no appreciable shift in the $I-V$ characteristics at zero-crossing has been observed.

The time-scales within which the conductance decays will depend on the diffusivity of Cu ions from the filament. However, the Cu diffusion coefficients reported from previous experiments are low enough to neglect any room temperature diffusion in the observed time scale.[58−60] Further, it is not well-established if the diffusion coefficient estimations account for other possible rate limiting steps such as initial Cu oxidation, Cu ion injection into SiO_2 dielectric, and the actual Cu ion migration. Also there are many orders of magnitude difference in the diffusion coefficient values reported due to the difference in experimental conditions such as microstructure and dimensions of deposited films and the experimental method (see Supporting Information Table S1). For example, Cu ion diffusion activation energy is shown to increase with increasing density of SiO_2.[61] Our devices have a 10 nm amorphous SiO_2 film, and the Cu atoms are already distributed within the SiO_2 film when the filament decay is observed. Attempts have been made to simulate isolated Cu ion diffusion in SiO_2 thin films. Guzman et al. report through DFT calculations that actual Cu ion migration barrier is in the range of 0.4−1.1 eV.[62] The diffusion coefficient estimated by Zeleny et al. through ab initio simulations for Cu ion diffusion in α-cristobalite crystalline form of SiO_2 is of the order of 10^{-8} cm^2/s (room temperature value determined from the given parameters[63] using the Arrhenius equation for diffusion coefficient) is much higher than the experimentally reported values.[58−60]

To study the dynamics of filament decay, we performed a 2D Cu diffusion simulation with initial filament structures of uniform concentration using COMSOL with Matlab. The physical radius of the conductive filament is Gaussian distributed along the x direction (in Figure 3f), with a mean radius of 2 nm and a standard deviation of 1 nm which is comparable to the estimated Cu filament radius. The interstitial diffusion of the Cu is modeled using Fick's law. In our simulation, the filament break time is determined as the time when the minimum concentration along the filament falls below 10% at some point in the y−axis. An example profile of initial random Cu filament structure and concentration at the filament break and at an intermediary time-step is shown in Figure 3f. A typical Cu concentration profile at different intermediary time-steps at the middle of the dielectric film is shown in Figure 3g. Figure 3h shows the distribution of simulated filament break times over 10^3 runs. We assumed the diffusion coefficient of Cu to be 5×10^{-16} cm^2/s, to obtain a match between the experimental and simulated mean time for the filament dissolution. This value is relatively close to the diffusion coefficient (2×10^{-13}cm^2/s) reported using cyclic voltammetry method.[28] Note that there was no appreciable change in the HRS of the uncycled devices over a few months, showing that there was no appreciable diffusion of Cu atoms from the electrodes into the SiO_2 at room temperature, which is in accordance with the earlier experimental results. However, for Cu filament already embedded in SiO_2, the formation energy barrier can be relatively lower, and a higher diffusion coefficient can be expected.

The observation of quantized conductance and filament radius calculation indicates the high scaling potential of the device. The material stack Cu/SiO_2/W also makes the device CMOS and BEOL compatible. However, the low power operation, which seems to have originated from the high mobility of Cu in SiO_2, also comes with high Cu diffusivity. The relatively lower retention we observed in the nanodimensional Cu filaments indicates a trade-off needed between operating voltage, material choice, device structure, and so forth.

In summary, we realized a memristive device with very low switching voltage (~250 mV) and operating currents (tens of μA) and demonstrated the quantized conductance nature of Cu filament based memristive device for the first time. The device showed conductance quantization at half-integer multiples of $2e^2/h$ and this behavior is explained mathematically by the presence of energy sub-bands in the Fermi level split between contact reservoirs. We showed that nanodimensional Cu filaments in SiO_2 undergoes spontaneous decay and the resulting conductance degradation behavior is modeled using diffusion simulations. Further, assuming a single filament, we used the numbers from our experimental observation to get an estimation of sub-band spacing, Cu filament radius, and Cu diffusion coefficient in SiO_2. Our work clearly illustrates the trade-offs involved in engineering low-power memristive devices. A highly mobile dopant in a dielectric could potentially give fast and low voltage switching. However, the programmed memory state in such devices may suffer from lower retention times, due to the diffusivity of the conductive species. Though such characteristics may not be ideal for engineering non-volatile memory devices, they are suited for mimicking short-term synaptic plasticity behavior of biological synapses for next-generation non-Von Neumann computing systems, where the acquired memory needs to be retained only for shorter periods of time.

■ ASSOCIATED CONTENT

Supporting Information

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Fabrication process, electrical and material characterization methods, and numerical simulation detail (PDF)

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Author Contributions

N.S.R. designed the experiment, fabricated the devices and characterized them, and wrote the manuscript. M.M., C.D., and S.N. performed the XPS and data analysis, and contributed to the manuscript. B.R. designed, planned, and supervised all phases of the project and edited the manuscript. All authors proofed the manuscript.

Notes

The authors declare no competing financial interest.

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